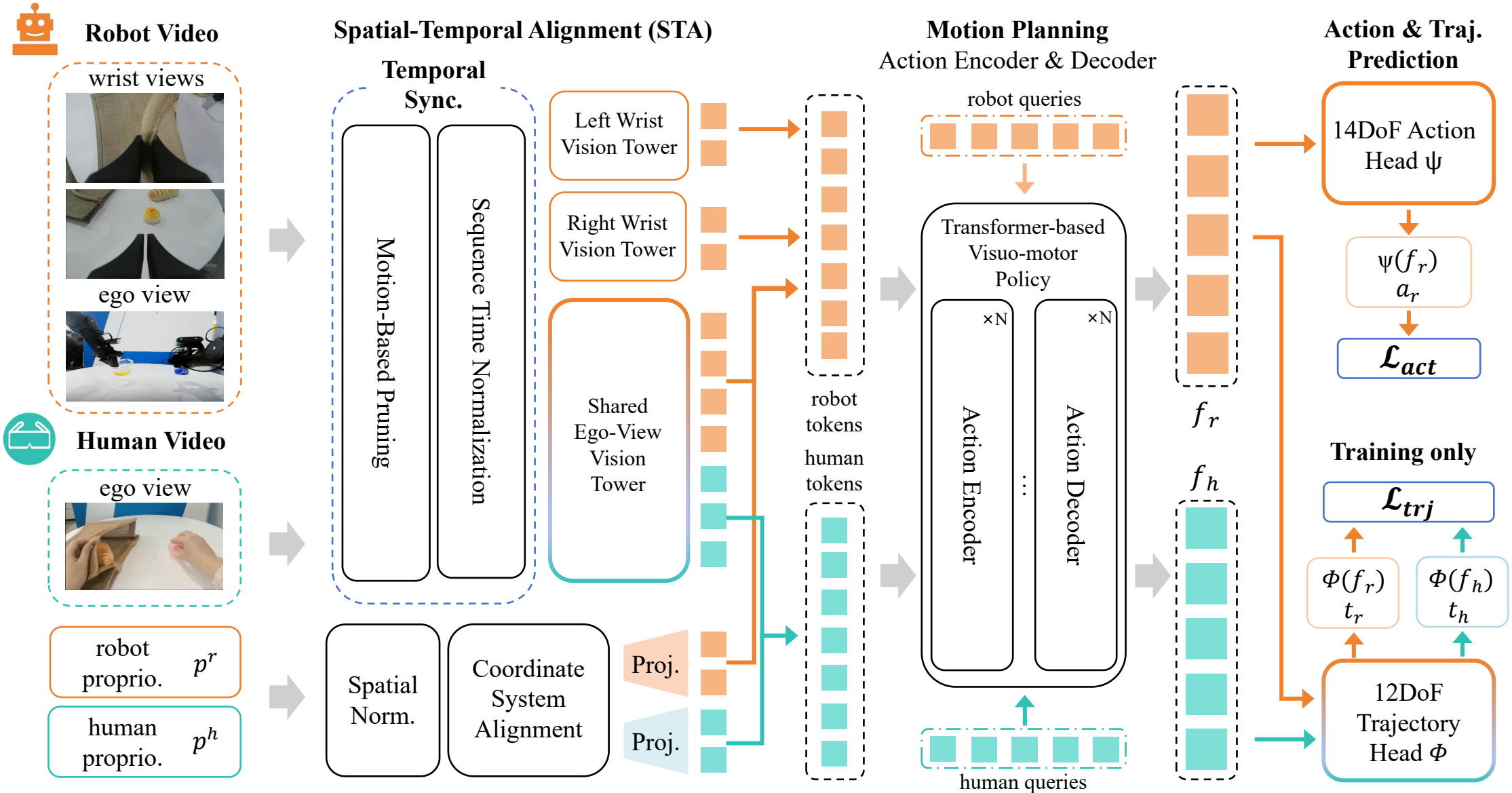
Robot Video
wrist views
ego view
Human Video
ego view
robot proprio. p^r
human proprio. p^h
Spatial-Temporal Alignment (STA)
Temporal Sync.
Motion-Based Pruning
Sequence Time Normalization
Spatial Norm.
Coordinate System Alignment
Left Wrist Vision Tower
Right Wrist Vision Tower
Shared Ego-View Vision Tower
Proj.
Proj.
robot tokens
human tokens
Motion Planning
Action Encoder & Decoder
robot queries
Transformer-based Visuo-motor Policy
×N
Action Encoder
×N
Action Decoder
human queries
f_r
f_h
Action & Traj. Prediction
14DoF Action Head ψ
$\psi(f_r)$
a_r
$\mathcal{L}_{act}$
Training only
$\mathcal{L}_{trj}$
$\Phi(f_r)$
t_r
$\Phi(f_h)$
t_h
12DoF Trajectory Head Φ